





ANALYZING REAL ESTATE MARKET WITH CAPM: A COMPARATIVE STUDY OF HOUSE PRICES AND RENTS IN MAJOR US CITIES.

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OBJECTIVE:

**TO FIND BEST CITIES FOR REAL
ESTATE INVESTMENT WITH HELP OF
CAPM.**





Capital asset pricing model

$$ER_i = R_f + \beta_i (ER_m - R_f)$$

Where:

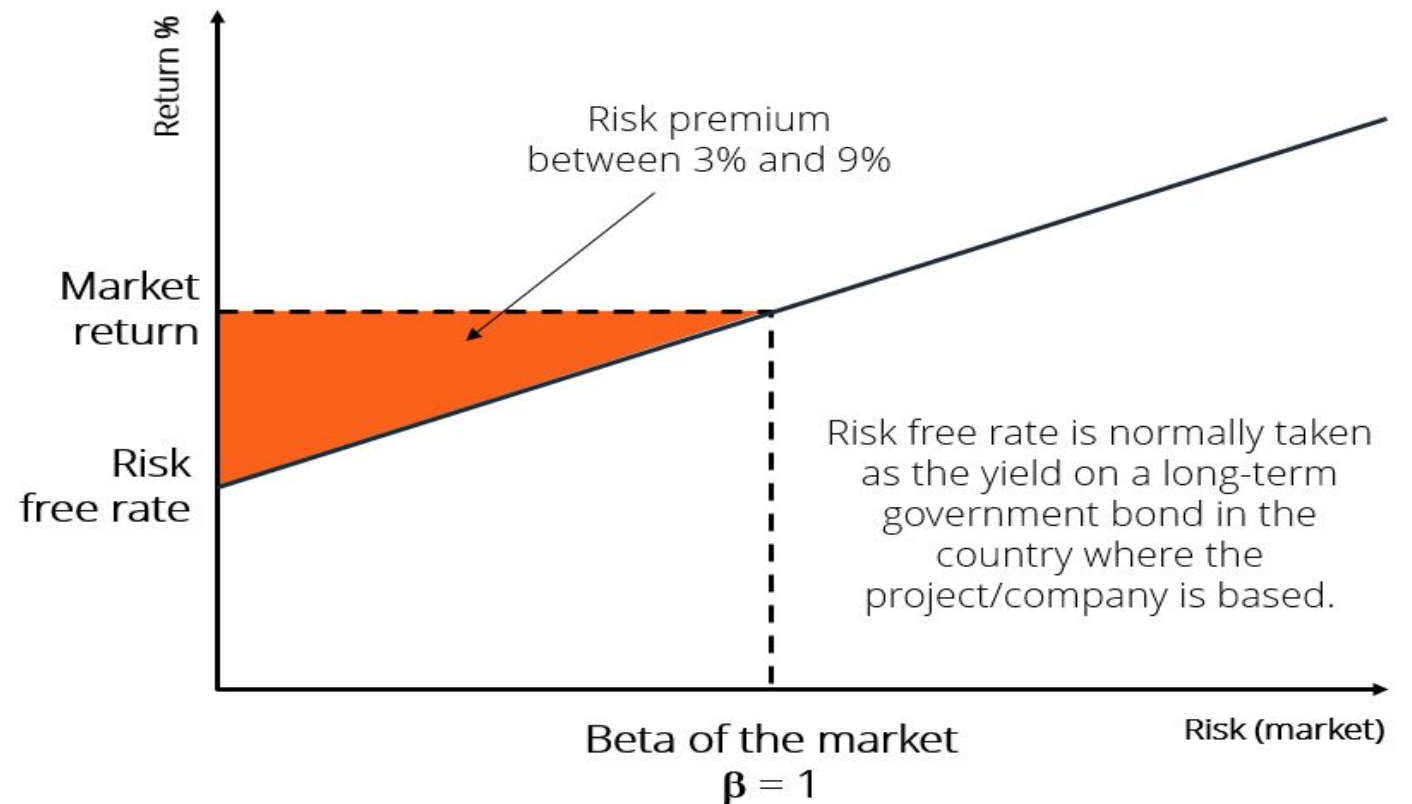
ER_i = expected return of investment

R_f = risk-free rate

β_i = beta of the investment

$(ER_m - R_f)$ = market risk premium

Capital Asset Pricing Model





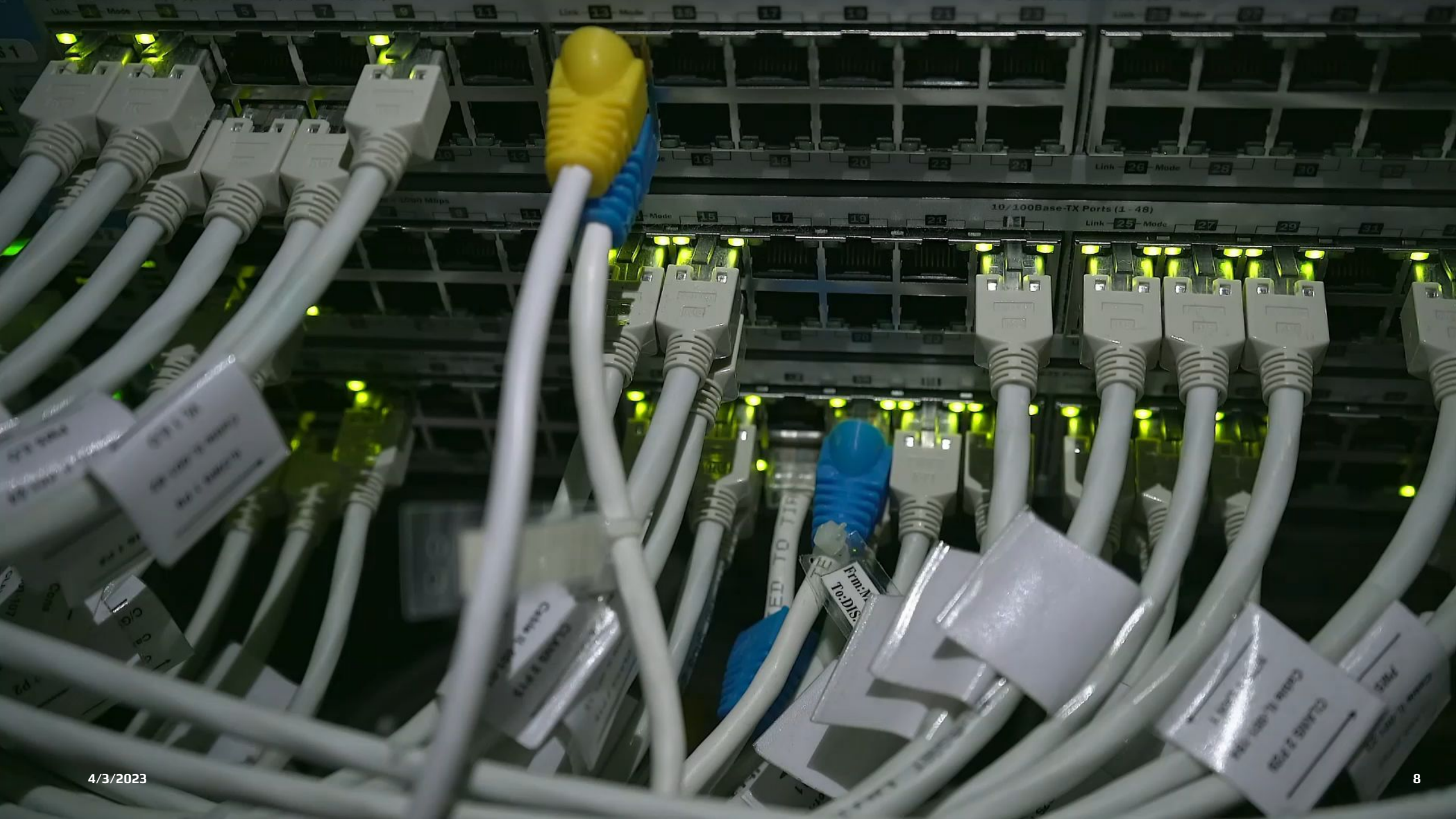
CAPM IN REAL ESTATE:

- To check property is fairly valued
- Expected Returns



- **Unique asset class**
- **Characteristics** -equity and fixed income securities.
- **Benefits**- rental income and capital appreciation.
- **Risks** -interest rate risk, market risk, and liquidity risk

WHY REAL ESTATE?



Zillow Observed Rent

Index:

- United States
- National, Metropolitan, County, City and Zip levels(510)
- From 03-31-2015 To 02-28-2023
- Lot of NAN values for Rent.

Zillow Home Value Index:

- United States
- Seven Geographic Levels(894)
- From 01-31-2000 To 02-28-2023
- Lot of NAN values for Rent.



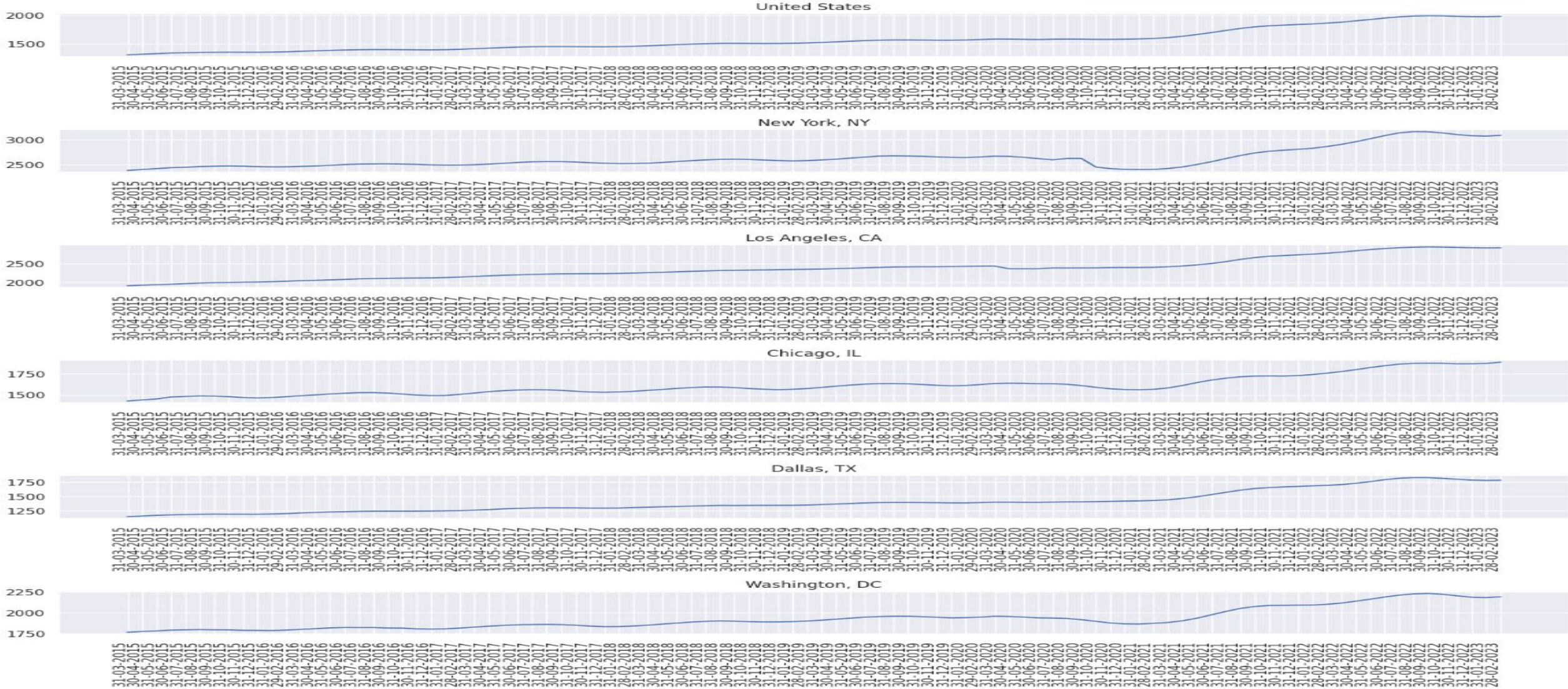
Zillow Observed Rent Index:

	RegionID	SizeRank	RegionName	RegionType	StateName	31-03-2015	30-04-2015	31-05-2015	30-06-2015	31-07-2015	...	31-05-2022	30-06-2022
0	102001	0	United States	country	NaN	1304.025668	1314.111855	1323.867218	1333.966852	1340.103382	...	1921.564887	1946.864887
1	394913	1	New York, NY	msa	NY	2386.585780	2405.273501	2422.713710	2440.732457	2449.784467	...	3017.668773	3079.744467
2	753899	2	Los Angeles, CA	msa	CA	1918.511039	1931.679906	1945.134903	1956.521807	1972.651475	...	2863.225552	2888.255552
3	394463	3	Chicago, IL	msa	IL	1427.222929	1440.010040	1451.347989	1475.071800	1481.594637	...	1825.495290	1846.855290
4	394514	4	Dallas, TX	msa	TX	1148.814439	1159.221531	1171.106575	1178.902425	1185.122950	...	1754.977334	1789.077334
...	...	...	...	...	...	...	...	...	...	...	...	...	...
506	753923	841	The Dalles, OR	msa	OR	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN
507	395110	843	Spearfish, SD	msa	SD	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN
508	394643	846	Great Bend, KS	msa	KS	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN
509	394791	849	Lexington, NE	msa	NE	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN
510	394718	892	Jamestown, ND	msa	ND	NaN	NaN	NaN	NaN	NaN	...	NaN	NaN

	United States	New York, NY	Los Angeles, CA	Chicago, IL	Dallas, TX	Houston, TX	Washington, DC	Miami, FL	Philadelphia, PA	Atlanta, GA	...	Breckenridge, CO	Sus
31-03-2015	1304.025668	2386.585780	1918.511039	1427.222929	1148.814439	1280.894885	1764.413107	1584.341775	1313.413202	1081.941328	...	NaN	
30-04-2015	1314.111855	2405.273501	1931.679906	1440.010040	1159.221531	1288.877292	1774.019710	1590.453333	1319.528465	1090.788976	...	NaN	
31-05-2015	1323.867218	2422.713710	1945.134903	1451.347989	1171.106575	1297.907743	1781.095735	1597.115761	1328.951455	1100.512226	...	NaN	
30-06-2015	1333.966852	2440.732457	1956.521807	1475.071800	1178.902425	1306.981089	1790.187373	1606.903829	1334.440417	1110.848409	...	NaN	
31-07-2015	1340.103382	2449.784467	1972.651475	1481.594637	1185.122950	1311.971157	1793.913217	1614.870361	1338.163512	1119.923744	...	NaN	
...	...	...	...	...	...	...	...	...	...	...	...	...	...
31-10-2022	1984.580174	3131.359562	2925.944799	1875.555990	1811.405487	1649.009218	2222.487492	2759.817006	1804.728006	1951.021545	...	NaN	
30-11-2022	1976.348669	3099.155451	2917.848297	1869.430226	1796.557207	1640.600126	2203.718782	2771.715495	1798.561630	1940.851131	...	NaN	
31-12-2022	1971.209604	3075.999271	2909.862264	1870.250969	1781.243863	1635.055768	2186.492244	2770.376072	1794.738058	1934.748984	...	NaN	
31-01-2023	1970.064413	3068.361445	2904.846076	1873.910478	1775.729839	1640.477920	2182.352622	2767.303387	1794.003361	1930.091416	...	3950.086553	
28-02-2023	1975.992304	3083.935335	2907.943492	1888.474027	1778.671326	1648.400288	2191.622038	2760.375870	1795.245625	1936.322326	...	3695.833333	

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Zillow Rent Index: 5 Major Cities And US

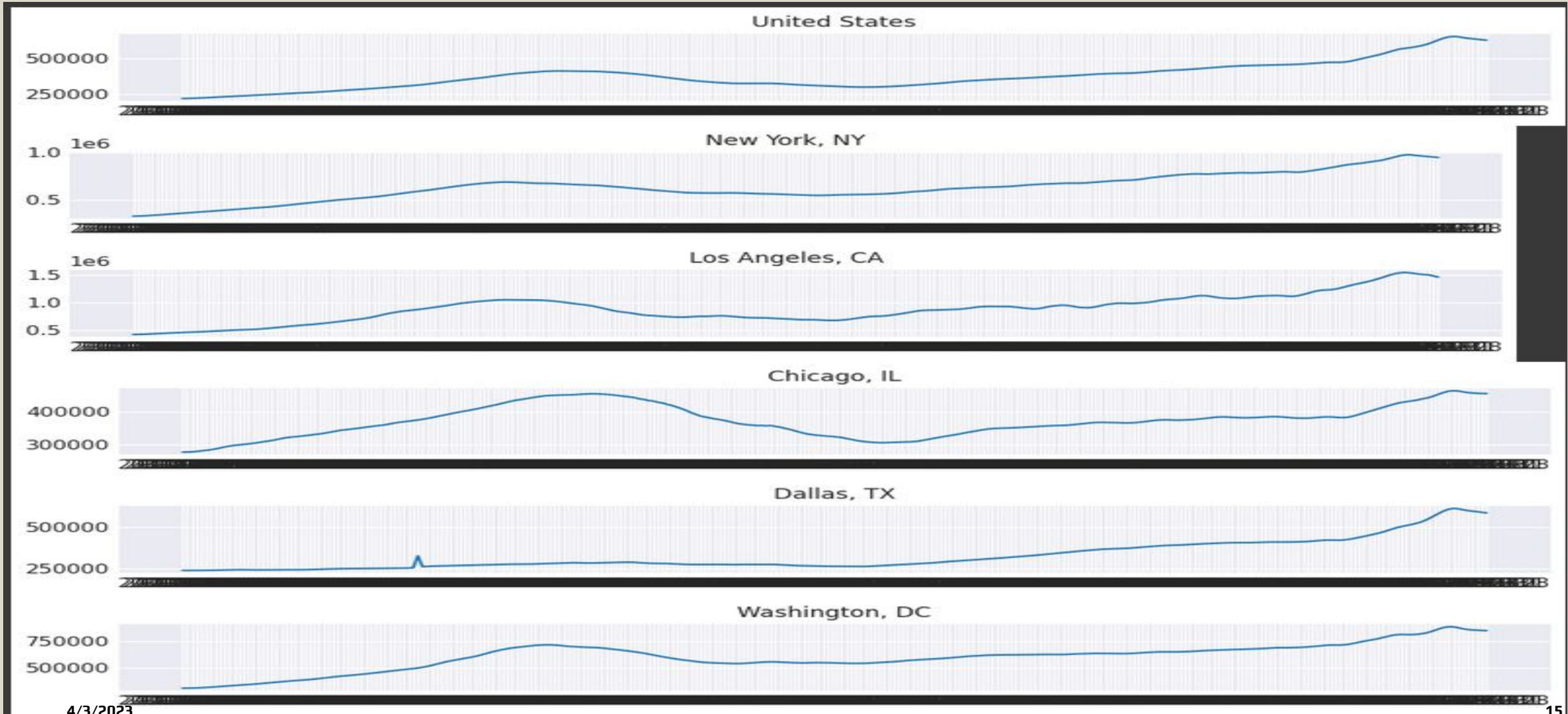


Zillow Home Value Index:

	RegionID	SizeRank	RegionName	RegionType	StateName	2000-01-31	2000-02-29	2000-03-31	2000-04-30	2000-05-31	...	2022-0
0	102001	0	United States	country	NaN	220956.345176	221458.998396	222066.241454	223359.090376	224825.147929	...	6.404198
1	394913	1	New York, NY	msa	NY	327452.287170	328852.051154	330251.730700	333105.250802	336042.253160	...	9.544795
2	753899	2	Los Angeles, CA	msa	CA	415704.708200	417370.538636	419437.299293	423600.150159	428055.156610	...	1.529614
3	394463	3	Chicago, IL	msa	IL	276086.448368	276429.409864	276995.934596	278409.083064	280055.910398	...	4.603203
4	394514	4	Dallas, TX	msa	TX	238197.499759	238217.611435	238281.255460	238476.462904	238768.997876	...	5.973016
...	...	...	...	...	...	...	...	...	...	...	...	...
891	753929	923	Zapata, TX	msa	TX	NaN	NaN	NaN	NaN	NaN	...	1.771712
892	394743	924	Ketchikan, AK	msa	AK	NaN	NaN	NaN	NaN	NaN	...	5.503749
893	753874	925	Craig, CO	msa	CO	171595.642405	171912.353882	172293.400410	172720.959377	173133.330084	...	3.754604
894	394767	926	Lamesa, TX	msa	TX	NaN	NaN	NaN	NaN	NaN	...	1.481475
895	395188	927	Vernon, TX	msa	TX	NaN	NaN	NaN	NaN	NaN	...	1.528707
896 rows × 283 columns												

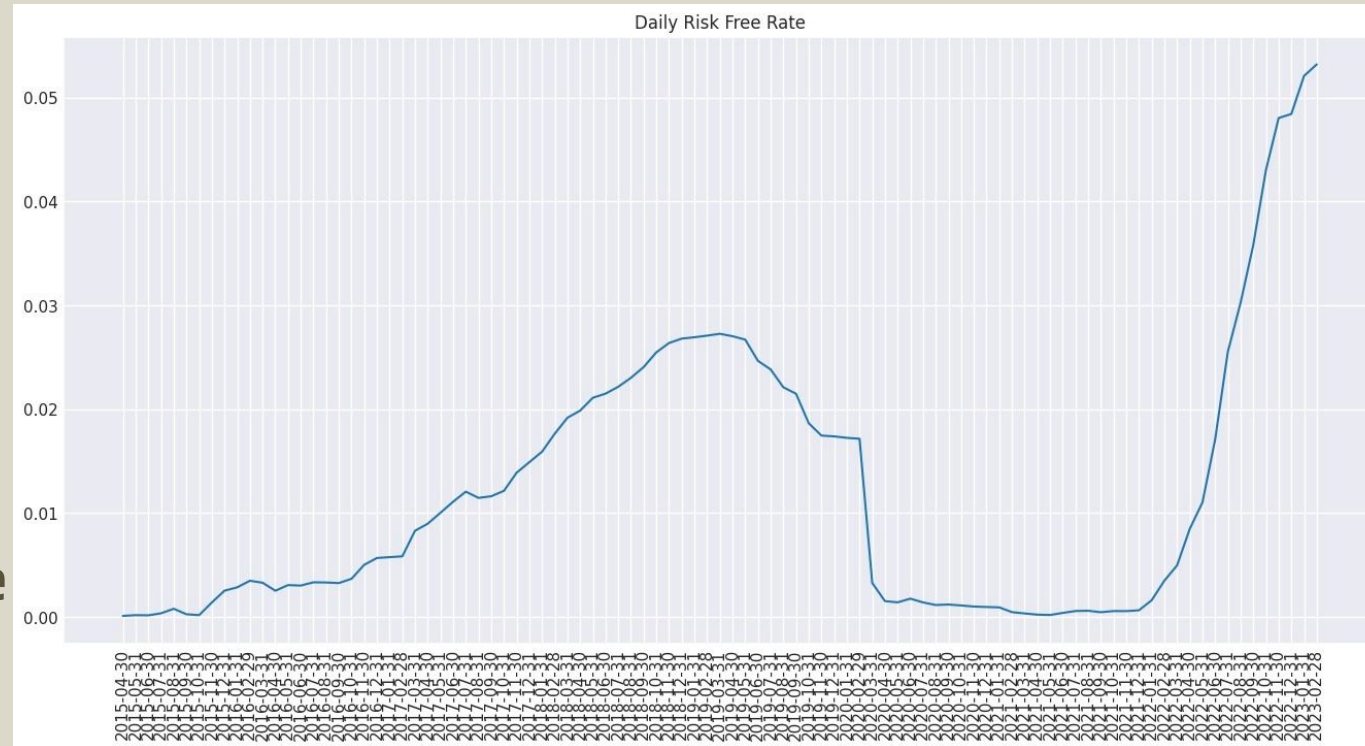
	United States	New York, NY	Los Angeles, CA	Chicago, IL	Dallas, TX	Houston, TX	Washington, DC	Miami, FL	Philadelphia, PA	Atlanta, GA
2000-01-31	220956.345176	327452.287170	4.157047e+05	276086.448368	238197.499759	222377.368105	302643.471536	226959.018927	210516.436973	262
2000-02-29	221458.998396	328852.051154	4.173705e+05	276429.409864	238217.611435	222458.557031	302997.300367	227598.095995	211060.245459	263
2000-03-31	222066.241454	330251.730700	4.194373e+05	276995.934596	238281.255460	222323.440860	303596.199943	228266.866083	211426.301573	263
2000-04-30	223359.090376	333105.250802	4.236002e+05	278409.083064	238476.462904	222480.951353	304944.200198	229610.681778	212342.606020	265
2000-05-31	224825.147929	336042.253160	4.280552e+05	280055.910398	238768.997876	222604.702697	306869.022723	230910.342216	213257.114134	267
...	...	...	...	...	...	...	...	...	...	...
2022-10-31	638961.423792	964539.973927	1.517686e+06	459378.184859	599258.914003	490407.316691	861344.424254	788242.295586	534005.921078	568
2022-11-30	635607.207309	960371.127039	1.510457e+06	458187.929753	595663.882752	489067.539862	857624.796365	787502.662760	534218.049173	566
2022-12-31	632254.337795	955474.805872	1.503368e+06	457111.141678	592026.563672	487258.889076	855122.494473	786625.787267	534600.005905	563
2023-01-31	628567.023664	950618.904060	1.485899e+06	456686.467974	588157.745238	485108.220828	852680.251379	783995.943751	534094.156713	561
2023-02-28	625420.530455	946658.178128	1.462508e+06	456436.476441	585114.644987	483874.660564	850767.085461	781365.872520	534151.496619	560
278 rows × 896 columns										

Zillow Home Value Index: 5 Major Cities And US

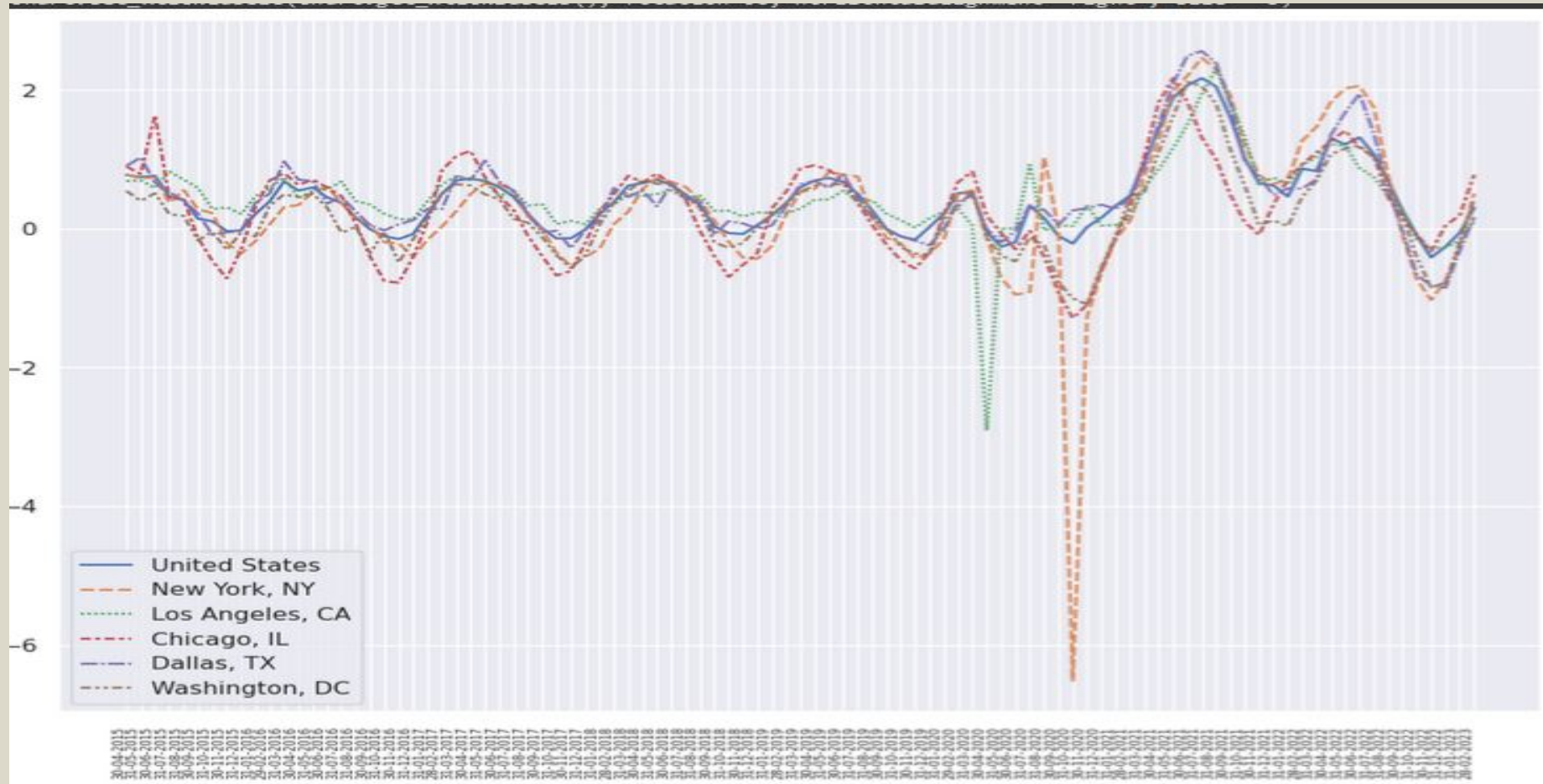


Risk Free Rate:

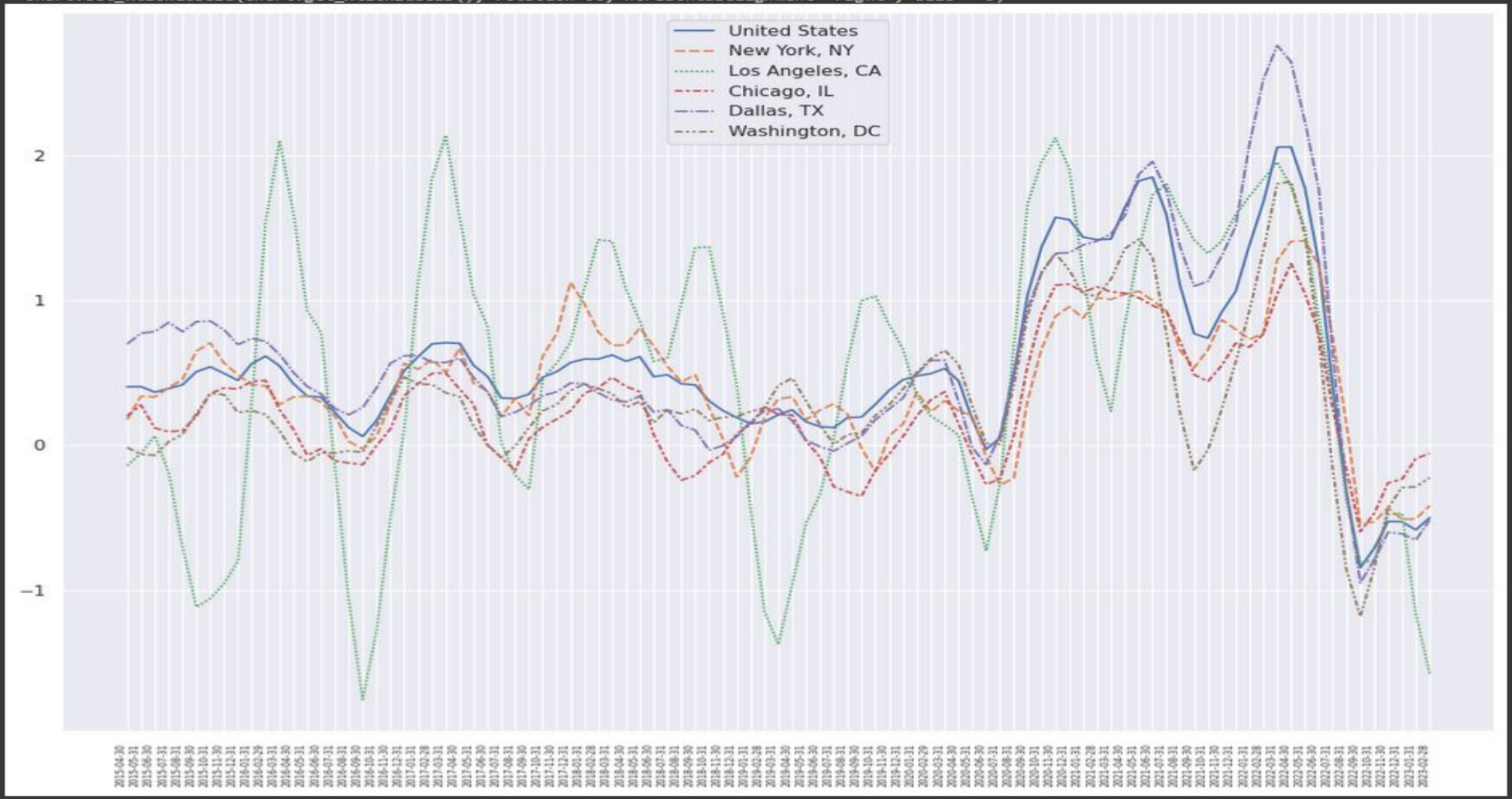
- From FRED API.
- Date: 31-04-2015 to 28-02-2023
- Converted Daily Risk Free Rate to Monthly Risk Free Rate (Considering mean.)



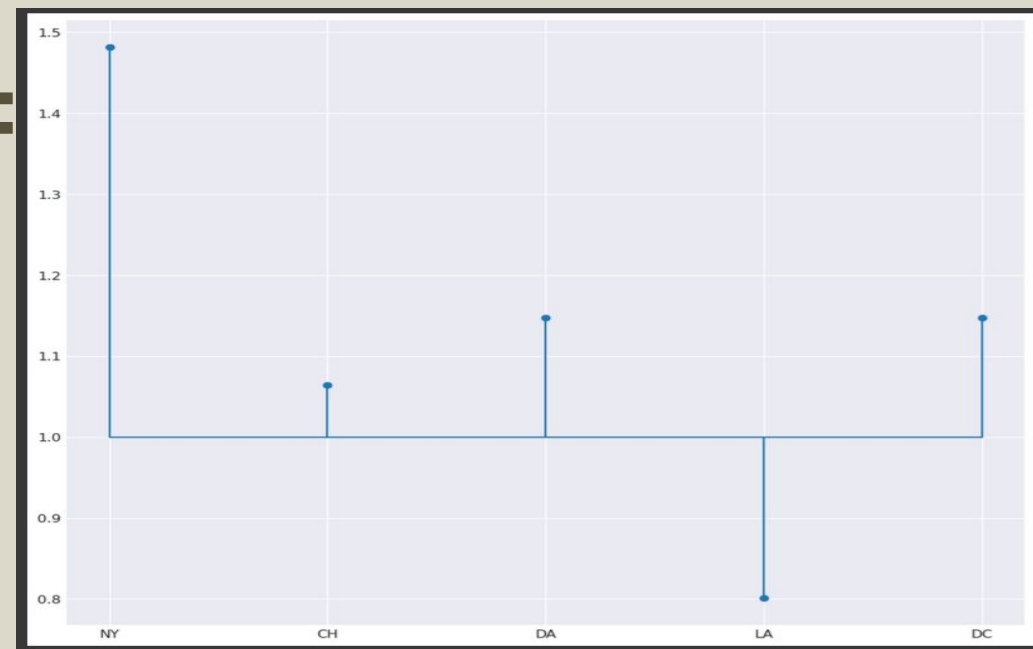
Excess Returns Rents:



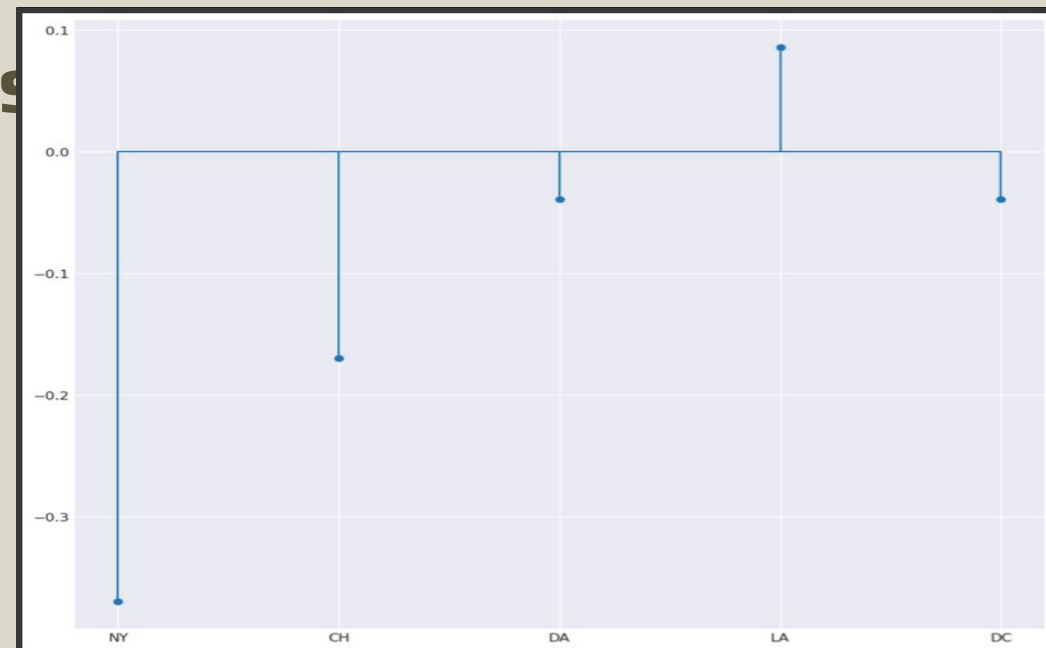
Excess Returns House Value



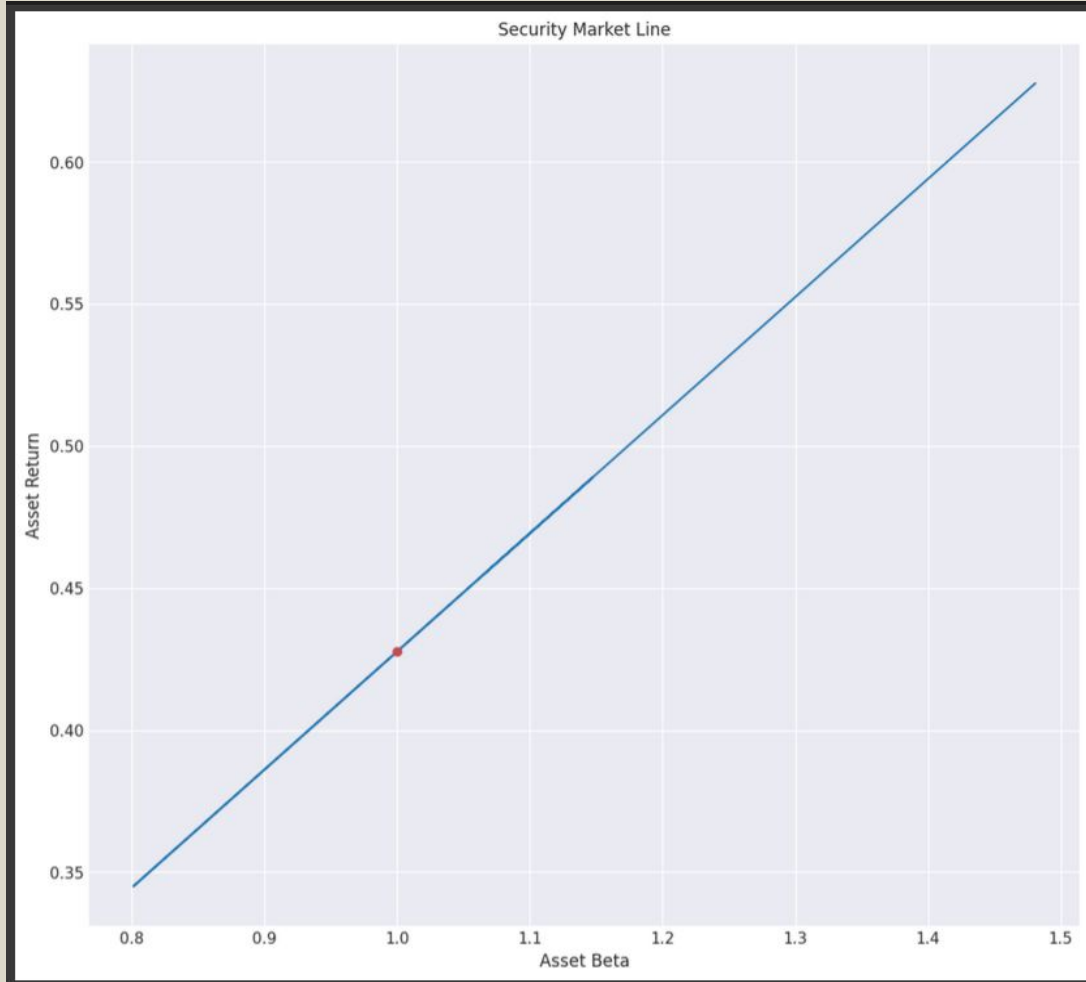
Computed Betas for Rents:



Computed Alphas for Rents

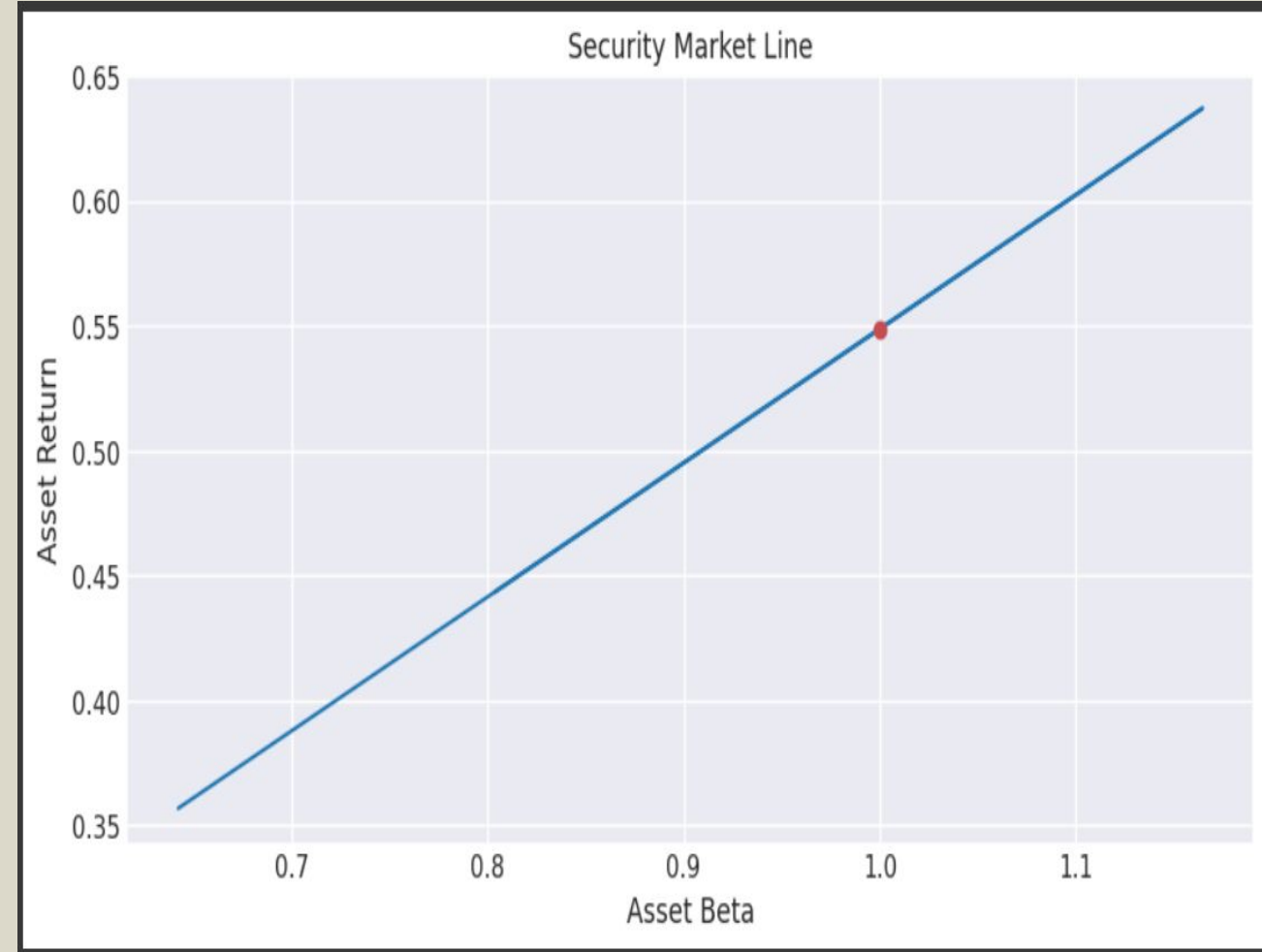


SML for Rents:



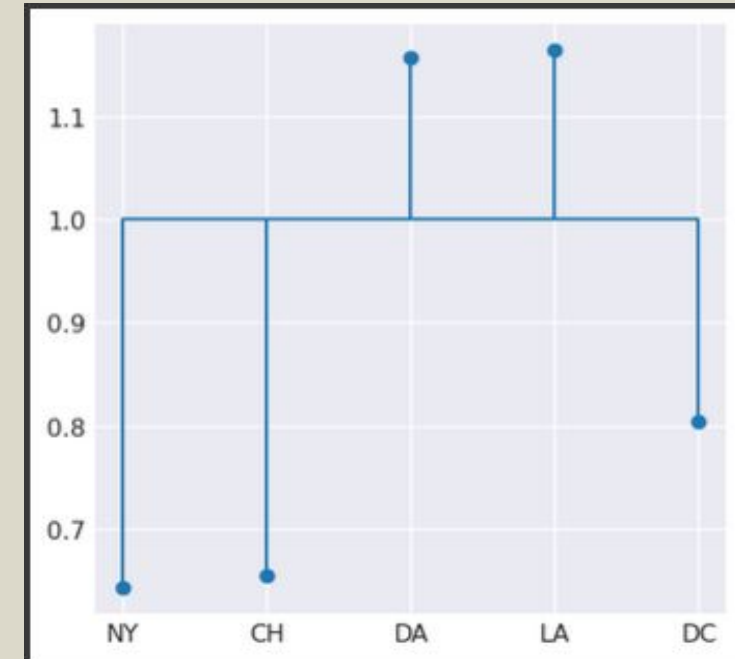
4/3/2023

SML for House Value:

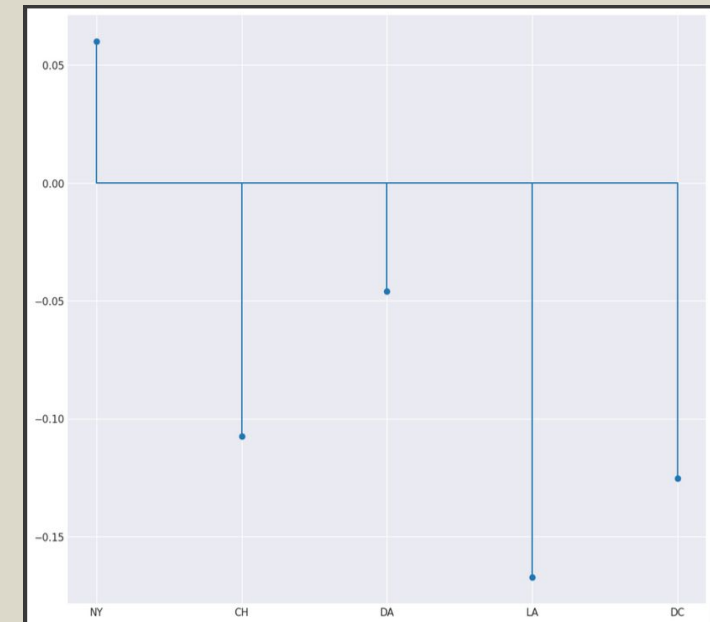


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Computed Betas for Homes:



Computed Alphas for Homes :



Concluding :

	Cities	Betas_Value	Alphas_Value	Betas_Rent	Alphas_Rent
0	NY	0.642181	0.059917	1.481016	-0.369840
1	CH	0.654374	-0.107423	1.064505	-0.169874
2	DA	1.157487	-0.046086	1.147098	-0.039550
3	LA	1.164700	-0.167299	0.801218	0.085485
4	DC	0.804130	-0.125450	1.147098	-0.039550



Thank You